Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/27/2006
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Project Title:	LOW VELOCITY FLOW BOILING OF ORDINARY LIQUIDS WITH EXTENSION TO LIQUID METALS- THE EFFECT OF REDUCED GRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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	Applications of boiling heat transfer in space can be found in the areas of thermal management, fluid handling and control, power systems, on-orbit storage and supply systems for cryogenic propellants and life support fluids, and for cooling of electronic packages for power systems associated with various instrumentation and control systems. Recent interest in exploration of Mars and other planets, and the concepts of in-situ resource utilization on Mars highlights the need to understand the effect of gravity on boiling heat transfer at gravity levels varying from 1 >= g/g_e >= 10^(-6). The objective of the proposed wok is to develop a mechanistic understanding of nucleate boiling and critical heat flux under low and micro-gravity conditions when the velocity of the imposed flow is small. For pool boiling, the effect of reduced gravity is to stretch both the length scale as well as the time scale for the boiling process. At high flow velocities, the inertia of the liquid determines the time and the length scales and as such the gravitational acceleration		

Task Description:	plays little role. However, at low velocities and at low gravity levels both liquid inertia and buoyancy are of equal importance. At present, we have little understanding of the interacting roles of gravity and liquid inertia on the nucleate boiling process. Little data that has been reported in the literature does not have much practical value in that it can not serve as a basis for design of heat exchange components to be used in space. The proposed work consists of both experimental and complete numerical simulations of the low velocity, low-gravity nucleate boiling process. A building block type of approach will be used in that first the growth and detachment process of a single bubble and flow and heat transfer associated with the sliding motion of the bubble over the heater surface after detachment will be studied. Thereafter, the complexity of the experiments with respect to the number of nucleation sites will be increased in steps. Liquid subcooling and flow velocity will be varied parametrically. The experiments are proposed to be conducted at 1 g_e, 10^(-2) g_e. The experiments at 10^(-2_ g_e will be conducted in the KC-135. In the laboratory experiments, particle velocimetry and holographic interferometry will be used to obtain data on velocity and temperature fields associated with a bubble prior to, and after detachment and during sliding motion. Numerical simulations modeling the micro and macro regions of the bubble will be carried out in three dimensions. Scaling the effect of gravity will be a prerequisite of the model. The results of the experiments will be used to validate analytical/numerical models. Task Objective: Task 1: Experiments at Earth Normal Gravity: The basic experiments with single, and merged bubbles attached to the heater surface and bubbles sliding along the heated wall will be carried out on surfaces inclined at different angles to the horizontal and subjected to forced flow parallel to the heater. The experiments will be conducted at one atmosphere pressure, but liquid subcooling and flow velocity will be varied parametrically. Water and PF-5060 will be used as test liquids. Task 2: Numerical Simulations/Models: In this task, transient temperature and flow field in vapor and liquid will be determined during growth of a single bubble subjected to flow parallel to the heater surface. Simulations will include the evaporation at the vapor-liquid interface and development of a microlayer underneath the bubble. This effort will be followed by simulations of situations in which neighboring bubbles merge, and bubbles slide along the heated surface for a long distance. The direction and magnitude of gravity, as well as flow velocity and subcooling will be variables of the problem. Task 3: Experiments in the KC-135 Aircraft: To understand bubble growth, detachment and subsequent motion of single and large merged bubbles, boiling experiments will be conducted under low gravity conditions (g/g_o 10^(-2)) of the aircraft. In these experiments, "designed" surfaces will be used. Visual observations and heat transfer data will be taken. The apparatus used in the laboratory experiments will also be employed with some modifications for experiments in the aircraft. The proposed research will provide fundamental understanding of bubble dynamics and heat transfer during nucleate boiling under low gravity and low velocity conditions. Such an understanding is needed for optimum design and safe operation of heat exchange equipment employing phase change for transfer of heat under microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Experiments Single bubble experiments have been perfomed at earth normal gravity and serve as the base data. These experiments were conducted for various liquid velocities, liquid subcoolings, wall superheats and orientations with respect to the gravity vector. Water was the test fluid. In order to investigate the effect of gravity on the bubble dynamics and associated heat transfer, single bubble experiments were also performed in the low gravity environment of KC-135. The experimental data show that the bubble size at sliding and lift-off increase as the gravity level decreases. Numerical Simulations Numerical simulations of single bubble nucleation, growth, sliding and departure have also been performed for the same test conditions using water and PF-5060. The bubble dynamics and the associated heat transfer have been quantified for various gravity levels, flow velocities, wall superheats, liquid subcoolings and orientations with respect to the gravity vector. The numerically predicted bubble size at departure does not match the experimentally measured bubble departure diameters.
Bibliography Type:	Description: (Last Updated: 08/23/2012)
Articles in Peer-reviewed Journals	Li D, Dhir VK. "Numerical study of a single bubble sliding on a downward facing heated surface." Journal of Heat Transfer. 2007 Jul;129:877-93. http://dx.doi.org/10.1115/1.2717943 , Jul-2007
Dissertations and Theses	Li D. "Numerical study of single bubble dynamics during flow boiling." Ph.D. Dissertation, University of California, Los Angeles, December 2005. , Dec-2005
Papers from Meeting Proceedings	Li D, Manickam S, Dhir VK. "A Numerical Study of a Single Bubble Sliding on a Downward Facing Heating Surface." ASME Summer Heat Transfer Conference, San Francisco, July 17–22, 2005. Proceedings of the ASME Summer Heat Transfer Conference, San Francisco, July 2005. Paper no. HT2005-72541; p. 171-178. http://dx.doi.org/10.1115/HT2005-72541 , Jul-2005
Papers from Meeting Proceedings	Li D, Dhir VK. "Single Bubble Dynamics during Flow Boiling on a Horizontal Surface at Different Gravity Levels." ASME Summer Heat Transfer Conference, San Francisco, CA, July 17–22, 2005. Proceedings of the ASME Summer Heat Transfer Conference, San Francisco, CA, July 2005. Paper no. HT2005-72852; p. 1035-1037. http://dx.doi.org/10.1115/HT2005-72852 , Jul-2005